Asssignment 1

Physics 443

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Question 1

Consider the wave function

$$\Psi(x,t) = Ae^{-\lambda|x|}e^{-i\omega t} \tag{1.1}$$

Where A, λ, ω are real and constant.

Question 1.a

Normalize Ψ and A. The answer can contain λ. [10 points]

Normalizing Ψ involves solving the following for A:

$$1 = \int_{-\infty}^{\infty} \Psi\Psi^* \,\mathrm{d}x = \int_{-\infty}^{\infty} A^2 e^{-2\lambda|x|} e^{i\omega t - i\omega t} \,\mathrm{d}x = \int_{-\infty}^{\infty} A^2 e^{-2\lambda|x|} \,\mathrm{d}x \tag{1.2}$$

Because this integral is symmetric about the y-axis, we can simply it to:

$$1 = 2A^2 \int_0^{\infty} e^{-2\lambda x} \,\mathrm{d}x \tag{1.3}$$

Trivially, this is:

$$1 = 2A^2 \left[\frac{1}{-2\lambda} e^{-2\lambda x} \right] \Bigg|_0^{\infty} = 2A^2 \left[0 - \frac{1}{-2\lambda}(1) \right] = \frac{A^2}{\lambda} \tag{1.4}$$

$$A = \sqrt{\lambda} \tag{1.5}$$

We then get:

$$\boxed{\Psi(x,t) = \sqrt{\lambda} e^{-\lambda|x|} e^{-i\omega t}} \tag{1.6}$$

Question 1.b

Determine the expectation values x and x^2. [20 points] What is the standard deviation of σ of x. the answers can contain λ. [15 points]

The expectation value x is given as:

$$\langle x \rangle = \int_{-\infty}^{\infty} \Psi(x,t) x \Psi^*(x,t)\,\mathrm{d}x \tag{1.7}$$

This is then:

$$\begin{aligned}
\int_{-\infty}^{\infty} \Psi(x,t) x \Psi^*(x,t)\,\mathrm{d}x &= \int_{-\infty}^{\infty} \sqrt{\lambda}^2 x e^{-\lambda|x|-\lambda|x|} e^{i\omega t - i\omega t}\,\mathrm{d}x && (1.8)\\
&= \int_{-\infty}^{\infty} \lambda x e^{-2\lambda|x|}\,\mathrm{d}x && (1.9)\\
&= \int_{0}^{\infty} \lambda x e^{-2\lambda x}\,\mathrm{d}x + \int_{-\infty}^{0} \lambda x e^{2\lambda x}\,\mathrm{d}x && (1.10)\\
&= \frac{\lambda}{(2\lambda)^2} + \lambda \left[\frac{1}{2\lambda} x e^{2\lambda x}\right]\Bigg|_{-\infty}^{0} - \lambda \int_{-\infty}^{0} \frac{1}{2\lambda} e^{2\lambda x}\,\mathrm{d}x && \text{(Hint 2 and IBP)}\\
&= \frac{1}{4\lambda} + 0 - \lambda \left[\frac{1}{(2\lambda)^2} e^{2\lambda x}\right]\Bigg|_{-\infty}^{0} && (1.11)\\
&= \frac{1}{4\lambda} - \frac{\lambda}{(2\lambda)^2} && (1.12)\\
&= \frac{1}{4\lambda} - \frac{1}{4\lambda} = 0 && (1.13)
\end{aligned}$$

So the expected value x is

$$\boxed{\langle x \rangle = 0} \tag{1.14}$$

The expectation value of x^2 is given as:

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \Psi(x,t) x^2 \Psi^*(x,t)\,\mathrm{d}x \tag{1.15}$$

Let's calculate it!

$$\begin{aligned}
\int_{-\infty}^{\infty} \Psi(x,t) x^2 \Psi^*(x,t)\,\mathrm{d}x &= \int_{-\infty}^{\infty} \sqrt{\lambda}^2 x^2 e^{-\lambda|x|-\lambda|x|} e^{i\omega t - i\omega t}\,\mathrm{d}x && (1.16)\\
&= \int_{-\infty}^{\infty} \lambda x^2 e^{-2\lambda|x|}\,\mathrm{d}x && (1.17)\\
&= \int_{0}^{\infty} \lambda x^2 e^{-2\lambda x}\,\mathrm{d}x + \int_{-\infty}^{0} \lambda x^2 e^{2\lambda x}\,\mathrm{d}x && (1.18)\\
&= \frac{2\lambda}{8\lambda^3} + ... && (1.19)
\end{aligned}$$

Now the integral on the right in equation 1.18 involves integration by parts twice, so instead I'll use the DI method as laid out below:

D		I
x^2	$+$	$\exp(2\lambda x)$
$2x$	$-$	$\dfrac{\exp(2\lambda x)}{2\lambda}$
2	$+$	$\dfrac{\exp(2\lambda x)}{4\lambda^2}$
0		$\dfrac{\exp(2\lambda x)}{8\lambda^3}$

This computes to

$$\left[x^2\frac{exp(2\lambda x)}{2\lambda} - 2x\frac{exp(2\lambda x)}{4\lambda^2} + 2\frac{exp(2\lambda x)}{8\lambda^3}\right]\Bigg|_{-\infty}^{0} = \frac{2}{8\lambda^3} \tag{1.20}$$

So we finally get:

$$\langle x^2\rangle = \frac{2\lambda}{8\lambda^3} + \frac{2\lambda}{8\lambda^3} = \frac{1}{2\lambda^2} \tag{1.21}$$

$$\boxed{\langle x^2\rangle = \frac{1}{2\lambda^2}} \tag{1.22}$$

Ermmmm lowkey did not clock that this function was symmetric lol. Now for the standard deviation, which is given as:

$$\sigma = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \boxed{\frac{\sqrt{2}}{2\lambda}} \tag{1.23}$$

Question 1.c

Below is a plot of $|\Psi(x,0)|^2$. We define a range of $[\langle x\rangle - \sigma, \langle x\rangle + \sigma]$. What is the probability of x being outside this range? [15 points] Can you mark out the $\langle x\rangle - \sigma$ and $\langle x\rangle + \sigma$ on the plot? (as vertical lines - the positions don't need to be accurate, and a rough sketch representing the correct positional relations will do) and mark out as *shadowed areas* the probability that x falls outside the range? [10 points]

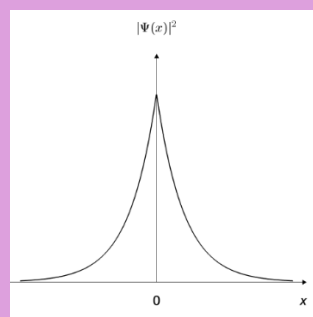

Figure 1: Plot of $|\Psi(x,0)|^2$ over x

What's nice is that the probability of x being outside the range, is that it is 1 minus the probability it is <u>within</u> the range. So we have:

$$P([\langle x\rangle - \sigma, \langle x\rangle + \sigma]) = 1 - \int_{-\sigma}^{\sigma} |\Psi(x,0)|^2 \,\mathrm{d}x \tag{1.24}$$

This integral:

$$\int_{-\sigma}^{\sigma} |\Psi(x,0)|^2 \,\mathrm{d}x = \int_{-\sigma}^{\sigma} \lambda e^{-2\lambda|x|} \,\mathrm{d}x \tag{1.25}$$

From our earlier results we get:

$$2\lambda\left[\frac{1}{-2\lambda}e^{-2\lambda x}\right]\Bigg|_0^{\sigma} = -\left[e^{-2\lambda\sigma} - 1\right] = 1 - e^{-2\lambda\sigma} \tag{1.26}$$

We get for the argument:

$$-2\lambda\sigma = -\frac{2\lambda\sqrt{2}}{2\lambda} = -\sqrt{2} \tag{1.27}$$

This gives the probability as:

$$\boxed{P([\langle x\rangle - \sigma, \langle x\rangle + \sigma]) = e^{-\sqrt{2}} \approx 0.24} \tag{1.28}$$

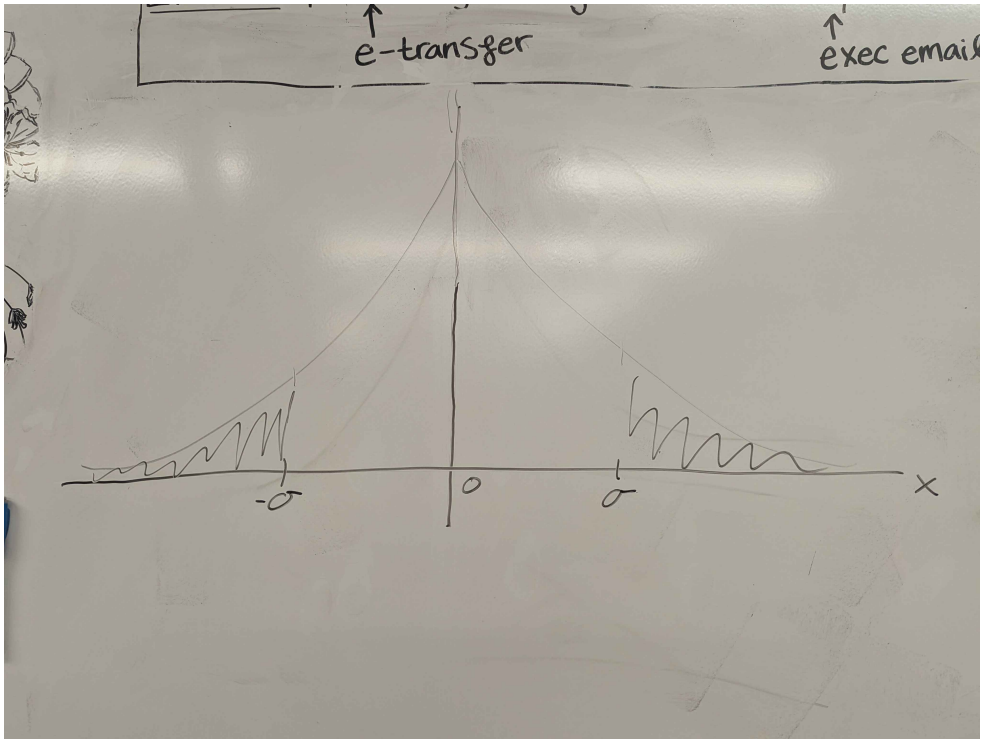

Figure 2: Filled in plot of figure 1

Question 2

In class, we derived $\frac{\partial\langle x\rangle}{\partial t} = \frac{\langle p\rangle}{m}$ using the definition of expectation values, the schrodinger equation, and integration by parts.

Question 2.a

Following a similar procedure, calculate $\frac{\partial\langle p\rangle}{\partial t}$ and (using the schrodinger equation) first show that

$$\frac{\partial\langle p\rangle}{\partial t} = \int_{-\infty}^{\infty}\left[-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi}{\partial x^2}\frac{\partial\Psi}{\partial x} - \Psi^*\frac{\partial^3\Psi}{\partial x^3}\right) - \Psi^*\frac{\partial V}{\partial x}\Psi\right]\mathrm{d}x \tag{2.1}$$

[15 points]

Let's start by getting to our definition of $\langle p\rangle$:

$$\langle p\rangle = -i\hbar\int_{-\infty}^{\infty}\Psi^*\frac{\partial\Psi}{\partial x} \tag{2.2}$$

Then we can take the partial w.r.t time:

$$\frac{\partial\langle p\rangle}{\partial t} = -i\hbar\int_{-\infty}^{\infty}\frac{\partial}{\partial t}\left(\Psi^*\frac{\partial\Psi}{\partial x}\right) \tag{2.3}$$

Using the product rule this partial in the right side becomes:

$$\frac{\partial}{\partial t}\left(\Psi^*\frac{\partial\Psi}{\partial x}\right) = \frac{\partial\Psi^*}{\partial t}\frac{\partial\Psi}{\partial x} + \Psi^*\frac{\partial}{\partial t}\frac{\partial\Psi}{\partial x} \tag{2.4}$$

$$= \frac{\partial\Psi^*}{\partial t}\frac{\partial\Psi}{\partial x} + \Psi^*\frac{\partial}{\partial x}\frac{\partial\Psi}{\partial t} \tag{2.5}$$

So now our partial momentum with time becomes:

$$\frac{\partial\langle p\rangle}{\partial t} = \int_{-\infty}^{\infty} -i\hbar\left(\frac{\partial\Psi^*}{\partial t}\frac{\partial\Psi}{\partial x} + \Psi^*\frac{\partial}{\partial x}\frac{\partial\Psi}{\partial t}\right) \tag{2.6}$$

$$= \int_{-\infty}^{\infty}\left(-i\hbar\frac{\partial\Psi^*}{\partial t}\right)\frac{\partial\Psi}{\partial x} + \frac{\partial}{\partial x}\left(-i\hbar\frac{\partial\Psi}{\partial t}\right)\Psi^* \tag{2.7}$$

$$= \int_{-\infty}^{\infty}\left(V\Psi^* - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Psi^*\right)\frac{\partial\Psi}{\partial x} + \frac{\partial}{\partial x}\left(\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Psi - V\Psi\right)\Psi^* \tag{2.8}$$

$$= \int_{-\infty}^{\infty} V\Psi^*\frac{\partial\Psi}{\partial x} - \frac{\hbar^2}{2m}\frac{\partial^2\Psi^*}{\partial x^2}\frac{\partial\Psi}{\partial x} + \frac{\hbar^2}{2m}\frac{\partial^3\Psi}{\partial x^3}\Psi^* - \Psi^*\frac{\partial V}{\partial x}\Psi - V\Psi^*\frac{\partial\Psi}{\partial x} \tag{2.9}$$

$$= \int_{-\infty}^{\infty}\left[-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi^*}{\partial x^2}\frac{\partial\Psi}{\partial x} - \Psi^*\frac{\partial^3\Psi}{\partial x^3}\right) - \Psi^*\frac{\partial V}{\partial x}\Psi\right]\mathrm{d}x \tag{2.10}$$

Question 2.b

Furthermore, you can use integration by parts to show that

$$\int_{-\infty}^{\infty} \left(\frac{\partial^2 \Psi^*}{\partial x^2} \frac{\partial \Psi}{\partial x} - \Psi^* \frac{\partial^3 \Psi}{\partial x^3} \right) \mathrm{d}x = 0 \tag{2.11}$$

Which means that the original equation $\frac{\partial \langle p \rangle}{\partial t} = \langle -\frac{\partial V}{\partial x} \rangle$? [15 points] Assume that Ψ and its derivatives vanish at $\pm\infty$, i.e infinitely far away.

Let's split this up into two integrals and then do integration by parts.

$$\int_{-\infty}^{\infty} \frac{\partial^2 \Psi^*}{\partial x^2} \frac{\partial \Psi}{\partial x} \, \mathrm{d}x - \int_{-\infty}^{\infty} \Psi^* \frac{\partial^3 \Psi}{\partial x^3} \, \mathrm{d}x \tag{2.12}$$

$$[...]\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial \Psi^*}{\partial x} \frac{\partial^2 \Psi}{\partial x^2} \, \mathrm{d}x - \int_{-\infty}^{\infty} \Psi^* \frac{\partial^3 \Psi}{\partial x^3} \, \mathrm{d}x \tag{2.13}$$

Since at infinity these quantities go to 0, when evaluated, we can just write these off. Let's expand the middle integral by parts again.

$$[...]\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \Psi^* \frac{\partial^3 \Psi}{\partial x^3} \, \mathrm{d}x - \int_{-\infty}^{\infty} \Psi^* \frac{\partial^3 \Psi}{\partial x^3} \, \mathrm{d}x \tag{2.14}$$

Oh wow this whole thing just goes to zero. Just as we expect.